AI log – Deliverable 2 - Ellen

During Deliverable 2, I used AI as a support tool to understand the assignment, develop the code step by step, and troubleshoot errors. I asked questions about Python functions, regular expressions, recursive file searching, reading files with different encodings, parsing log files, and creating and inspecting Pandas DataFrames.

Examples of questions included:

- How can I process log files and collect valid transactions into a list of dictionaries?
- How can I convert a list of dictionaries into a Pandas DataFrame and use timestamps as the index?
- What do error messages such as `TypeError` and `NameError` mean, and how can I fix them?
- How can I interpret the output from `df.info()` and check whether all transactions have been included?

My working method involved **trial and error**. I tested the code, examined the output, and used error messages to identify mistakes. For example, I corrected an incorrect reference to `pd` instead of `df`, and investigated why the DataFrame initially appeared to contain transactions from only one register. By printing register IDs for individual log files and checking the combined list, I confirmed that all nine files were processed correctly and that the DataFrame contained 720 transactions, with 240 transactions from each of the three registers.

AI was also used to explain code in simpler terms, improve comments, organise the code, and write short explanations for the notebook report. I reviewed the suggestions, tested changes in my own environment, and checked the results before continuing. This helped me improve my understanding of the code and solve problems systematically rather than simply copying solutions.

Use of AI – Gina

Questions Asked

1. Can you use .append() on an array in Python?
2. What is the code for calculating the standard deviation of a list?
3. Does this code work for finding/replacing values that are more than three standard deviations above the mean?
4. Can you use .append() on a NumPy array?

Summary (approx. 100 words):

We began by discussing the difference between standard Python lists and NumPy arrays. I explained that Python lists have the .append() method, while NumPy arrays do not, and that np.append() must be used instead. Unlike list .append(), np.append() returns a new array rather than modifying the existing one in place. We then looked at how to calculate standard deviation using either np.std() from NumPy or the built-in statistics library. Finally, we reviewed your outlier-handling code. I explained that == compares values rather than

assigning new ones, and that you must either iterate over indices or use NumPy masking to modify array elements directly.

Questions Asked

1. Whether Tasks 3 and 4 should use the output from a partner's Tasks 1 and 2, or work directly from the raw files.
2. Why the cursor in VS Code suddenly highlights an entire character instead of appearing as a vertical line.
3. What the refund ratio task in Task 3 actually means.
4. How to convert log entries into dictionaries.
5. Help with correctly formatting a list of dictionaries.
6. An error involving a missing argument in a function call to values_from_status.
7. Whether refund ratio should be calculated per product based on the number of sales and refunds.
8. Translation of the assignment text about grouping and resampling data in Pandas.

Summary (approx. 100 words):

I explained that Tasks 3 and 4 should generally be built around the data structures produced in Tasks 1 and 2, although you can use your own sample data while developing and testing. For VS Code, I identified the issue as most likely being caused by overwrite mode being enabled. We also discussed the refund ratio assignment and concluded that it most likely refers to the ratio of refunds to sales for each product, rather than a ratio based on monetary amounts. In addition, I helped convert log entries into Python dictionaries, correct formatting errors in lists of dictionaries, identify the cause of a TypeError in a function call, and translate the Pandas grouping and resampling assignment into Norwegian.

Questions Asked

1. Why does groupby("item").resample("1h").sum() produce many empty Sandwich rows?
2. How can I create a pivot table that counts the occurrences of each status per item?
3. How do I count how many times each status occurs for each item?
4. How can I extract the number of SUCCESS transactions per item into a list?
5. Why does counts["SUCCESS"] not work after groupby(...).size()?
6. Is the difference between the MultiIndex result and unstack() only the presentation?
7. How can I get both an array of SUCCESS counts and an array of REFUND counts per item?
8. How can I solve the task of cleaning outliers from a DataFrame containing prices?

Summary (50 words):

We reviewed grouping, resampling, and counting transaction statuses using groupby, crosstab, and unstack. We discussed the difference between MultiIndex Series and

DataFrames, how to extract SUCCESS and REFUND counts, and how to convert the results into arrays. Finally, we covered identifying price outliers and replacing them with the median of valid values.

Your Questions

1. In an assignment where a function in Task 2 returns a list of Path objects, is this the correct code for creating a list of tuples where the first element is all text from the file and the second element is the encoding?
2. This is the assignment I am working on, and I am currently on Task 5.
3. Thanks, but I only want hints for the tasks so that I have to think for myself as well.
4. If my partner and I split Task 5 so that she completes Tasks 1–3 and I complete Tasks 4–7, can we work simultaneously?
5. Could you create a dummy version of the output I need from her so I can develop my part?
6. In addition, output files such as cleaned_transactions.csv must be generated. Do I then need functions from Task 1 or Task 2?

Summary (approx. 50 words):

I explained that your code for collecting file contents and encodings into a list of tuples was correct. For Task 5, I suggested thinking of the workflow as a chain of functions. I showed how you and your partner could work in parallel using a dummy DataFrame and explained that cleaned_transactions.csv can be generated directly from the DataFrame without depending on Tasks 1 and 2 during development.